

AI Programming Assistant (AIPA)
Final Practical Assessment

Time: 3 Hours
Maximum Marks: 60

Instructions:

1. Attempt all questions.
2. Write clean and well-documented code.
3. Display outputs wherever applicable.
4. Use proper naming conventions and comments.
5. For ML/DL tasks, use the Heart Failure Prediction Dataset.

Dataset:

Heart Failure Prediction Dataset

<https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction>

Question 1: Python Programming & OOP (15 Marks)

Create a Hospital Management System using Python.

Requirements:

1. Create a class Patient with:
 - patient_id
 - patient_name
 - age
 - disease
2. Use a constructor to initialize values.
3. Create a method display_details() to display patient information.
4. Create at least 5 Patient objects and store them in a list.
5. Using loops:
 - Display all patient records.
 - Count patients whose age is greater than 50.
 - Display patients suffering from "Heart Disease".
6. Implement exception handling while taking user input for age.

Expected Concepts:

Classes, Objects, Constructor, Lists, Loops, Conditional Statements, Exception Handling

Question 2: MySQL Database Management (15 Marks)

Create a database named HospitalDB.

Task 1:

Create a table named Patients with the following fields:

- PatientID
- PatientName
- Gender
- Age
- Disease
- City
- BillAmount

Task 2:

Insert at least 10 records.

Task 3:

Write SQL queries to perform the following:

1. Display all patient records.
2. Display patients whose age is greater than 50.
3. Display patients suffering from Heart Disease.
4. Find the highest bill amount.
5. Find the average bill amount.
6. Count patients city-wise.
7. Display records sorted by BillAmount in descending order.
8. Display patients whose bill amount is above the average bill amount.
9. Create another table Doctors and perform an INNER JOIN.
10. Display the city having the highest number of patients.

Expected Concepts:

DDL, DML, Aggregate Functions, WHERE, ORDER BY, GROUP BY, HAVING, JOIN

Question 3: Data Science & Machine Learning (15 Marks)

Use the Heart Failure Prediction Dataset.

Task 1: Data Loading & Understanding

1. Load the dataset using Pandas.
2. Display:
 - Shape
 - Data Types
 - Missing Values
 - Statistical Summary

Task 2: Data Cleaning

1. Check for missing values.
2. Remove duplicate records.
3. Encode categorical variables if required.

Task 3: Exploratory Data Analysis

1. Plot:
 - Age Distribution
 - Heart Disease Distribution
 - Gender vs Heart Disease
 - Chest Pain Type Distribution
2. Create a correlation heatmap.

Task 4: Machine Learning

1. Define:
 - Features (X)
 - Target Variable (y)
2. Perform Train-Test Split (80:20).
3. Train the following models:
 - Logistic Regression
 - Decision Tree Classifier
4. Evaluate both models using:
 - Accuracy
 - Confusion Matrix
 - Classification Report
5. Compare both models and identify the better-performing model.

Expected Concepts:

Pandas, NumPy, Matplotlib, Seaborn, EDA, Train-Test Split, Classification Models, Evaluation Metrics

Question 4: Deep Learning & NLP (15 Marks)

Part A: Deep Learning

Using the Heart Failure Prediction Dataset:

1. Preprocess the dataset.
2. Split into Training and Testing datasets.
3. Create a Deep Neural Network using TensorFlow/Keras:

Architecture:

Input Layer

↓

Dense(16, activation='relu')

↓

Dense(8, activation='relu')

↓

Dense(1, activation='sigmoid')

4. Train the model for at least 20 epochs.
5. Evaluate:
 - Loss
 - Accuracy
6. Compare Deep Learning model accuracy with Machine Learning models from Question 3.

Part B: NLP

Perform the following operations on the sentence:

"Artificial Intelligence is transforming healthcare and improving patient outcomes."

1. Tokenization
2. Stopword Removal
3. Stemming
4. POS Tagging
5. Named Entity Recognition (NER) using spaCy
6. Perform Sentiment Analysis using TextBlob

Display outputs for each step.

Expected Concepts:

TensorFlow, Keras, Dense Layers, Model Training, NLP Pipeline, NLTK, spaCy, TextBlob